

## ARTICLE KD-2

### BASIC DESIGN REQUIREMENTS

#### KD-200 SCOPE

This Article provides basic design rules and definitions for vessels constructed in accordance with this Code. Additional rules for fatigue life and special construction techniques are given in later Articles of this Part.

If construction details do not satisfy the various configurations contained herein, or if no applicable equations are presented, a detailed stress analysis shall be made to show conformance with this Part.

#### KD-210 TERMS RELATING TO STRESS ANALYSIS

Terms used relating to stress analysis are defined below.

(a) *Autofrettage*. Autofrettage is a process for producing a system of favorable residual stresses in a thick walled vessel by pressurizing to produce plastic deformation in part or all of the wall thickness.

(b) *Deformation*. Deformation of a component part is alteration of its shape or size due to stress or temperature changes.

(c) *Fatigue Strength Reduction Factor*. This is a stress intensification factor which accounts for the effect of a local structural discontinuity (stress concentration) on the fatigue strength. Values for some specific cases, based on experiment, are given elsewhere in this Division. In the absence of experimental data, the theoretical stress concentration factor may be used.

(d) *Gross Structural Discontinuity*. A gross structural discontinuity is a source of stress or strain intensification which affects a relatively large portion of a structure and has a significant effect on the overall stress or strain pattern or on the structure as a whole. Examples of gross structural discontinuities are head-to-shell and flange-to-shell junctions, nozzles, and junctions between shells of different diameters or thicknesses.

(e) *Inelasticity*. Inelasticity is a general characteristic of material behavior in which the material does not return to its original (undeformed) shape and size after

removal of all applied loads. Plasticity and creep are special cases of inelasticity.

(1) *Plasticity*. Plasticity is the special case of inelasticity in which the material undergoes time-independent nonrecoverable deformation.

(2) *Plastic Analysis*. Plastic analysis is that method which computes the structural behavior under given loads considering the plasticity characteristics of the materials including strain hardening and the stress redistribution occurring in the structure. (Strain rate effects may also be significant where impact or other dynamic loads are involved.)

(3) *Plastic Instability Load*. The plastic instability load for members under predominantly tensile or compressive loading is defined as that load at which unbounded plastic deformation can occur without an increase in load. At the plastic tensile instability load, the true stress in the material increases faster than strain hardening can accommodate.

(4) *Strain Limiting Load*. When a limit is placed upon a strain, the load associated with the strain limit is called the strain limiting load.

(5) *Limit Analysis*. Limit analysis is a special case of plastic analysis in which the material is assumed to be ideally plastic (nonstrain-hardening). In limit analysis, the equilibrium and flow characteristics at the limit state are used to calculate the collapse load. Two bounding methods are used in limit analysis: the lower bound approach, which is associated with a statically admissible stress field, and the upper bound approach, which is associated with a kinematically admissible velocity field. For beams and frames, the term *mechanism* is commonly used in lieu of *kinematically admissible velocity field*.

(6) *Collapse Load — Limit Analysis*. The methods of limit analysis are used to compute the maximum load a structure made of ideally plastic material can carry. The deformations of an ideally plastic structure increase without bound at this load, which is termed collapse load.

(7) *Plastic Hinge*. A plastic hinge is an idealized concept used in limit analysis. In a beam or frame, a plastic hinge is formed at the point where the moment, shear, and axial force lie on the yield interaction surface. In plates and shells, a plastic hinge is formed where the generalized stresses lie on the yield surface.

(8) *Creep*. Creep is the special case of inelasticity that relates to the stress-induced time-dependent deformation under load. Small time-dependent deformations may occur after the removal of all applied loads.

(9) *Ratcheting*. Ratcheting is a progressive incremental inelastic deformation or strain which can occur in a component that is subjected to variations of mechanical stress, thermal stress, or both (thermal stress ratcheting is partly or wholly caused by thermal stress).

(10) *Shakedown*. Shakedown of a structure occurs if, after a few cycles of load application, ratcheting ceases. The subsequent structural response is elastic, or elastic-plastic, and progressive incremental inelastic deformation is absent. Elastic shakedown is the case in which the subsequent response is elastic.

(11) *Free End Displacement*. Free end displacement consists of the relative motions that would occur between an attachment and connected structure or equipment if the two members were separated. Examples of such motions are those that would occur because of relative thermal expansion of piping, equipment, and equipment supports, or because of rotations imposed upon the equipment by sources other than the piping.

(12) *Expansion Stresses*. Expansion stresses are those stresses resulting from restraint of free end displacement.

(f) *Load Stress*. The stress resulting from the application of a load, such as internal pressure or the effects of gravity, as distinguished from thermal stress.

(g) *Local Primary Membrane Stress  $P_L$* . Cases arise in which a membrane stress produced by pressure or other mechanical loading and associated with a primary loading, discontinuity, or both effects would, if not limited, produce excessive distortion in the transfer of load to other portions of the structure. Conservatism requires that such a stress be classified as a local primary membrane stress even though it has some characteristics of a secondary stress. An example of a local primary membrane stress is the membrane stress in a shell produced by external load and moment at a permanent support or at a nozzle connection.

(h) *Local Structural Discontinuity*. A local structural discontinuity is a source of stress or strain intensification that affects a relatively small volume of material and does not have a significant effect on the overall stress

or strain pattern or on the structure as a whole. Examples are small fillet radii and small attachments.

(i) *Membrane Stress*. Membrane stress is the component of normal stress that is uniformly distributed and equal to the average value of stress across the thickness of the section under consideration.

(j) *Normal Stress  $\sigma$* . The component of stress normal to the plane of reference (this is also referred to as *direct stress*). Usually, the distribution of normal stress is not uniform through the thickness of a part, so this stress is considered to be made up in turn of two components, one of which is uniformly distributed and equal to the average value of stress across the thickness of the section under consideration, and the other of which varies with the location across the thickness.

(k) *Operational Cycle*. An operational cycle is defined as the initiation and establishment of new conditions followed by a return to the conditions that prevailed at the beginning of the cycle. Three types of operational cycles are considered:

(1) start-up/shutdown cycle, defined as any cycle that has atmospheric temperature, pressure, or both as its extremes and normal operation conditions as its other extreme;

(2) the initiation of and recovery from any emergency or upset condition that shall be considered in the design;

(3) normal operating cycle, defined as any cycle between start-up and shutdown which is required for the vessel to perform its intended purpose.

(l) *Peak Stress  $F$* . The basic characteristic of a peak stress is that it does not cause any noticeable distortion and is objectionable only as a possible source of a fatigue crack or a brittle fracture. A stress that is not highly localized falls into this category if it is of a type which cannot cause progressive deformation (ratcheting). Examples of peak stress are:

(1) the thermal stress which occurs when a relatively thin inner shell material is dissimilar from a relatively thick outer shell material;

(2) the thermal stress in the wall of a vessel or pipe caused by a rapid change in temperature of the contained fluid;

(3) the stress at a local structural discontinuity.

(m) *Primary Stress*. A normal stress or a shear stress developed by the imposed loading which is necessary to satisfy the simple laws of equilibrium of external and internal forces and moments. The basic characteristic of a primary stress is that it is not self-limiting. Primary stresses that considerably exceed the yield strength will result in failure or at least in gross distortion. A thermal stress is not classified as a primary stress. Primary

membrane stress is divided into general and local categories. A general primary membrane stress is one which is so distributed in the structure that no redistribution of load occurs as a result of yielding. Examples of primary stress are:

(1) average through-wall longitudinal stress and the average through-wall circumferential stress in a closed cylinder under internal pressure, remote from discontinuities;

(2) bending stress in the central portion of a flat head due to pressure.

(n) *Secondary Stress*. A secondary stress is a normal stress or a shear stress developed by the constraint of adjacent parts or by self-constraint of a structure. The basic characteristic of a secondary stress is that it is self-limiting. Local yielding and minor distortions can satisfy the conditions that cause the stress to occur and failure from one application of the stress is not to be expected. Examples of secondary stress are:

(1) general thermal stress [see KD-210(l)(1)];

(2) bending stress at a gross structural discontinuity.

(o) *Shape Factor*. The shape factor  $\alpha$  is defined as the ratio of the moment that produces a full plastic section (no strain hardening) to the bending moment that produces initial yielding at the extreme fibers of the section.  $\alpha = 1.5$  for a rectangular section.

(p) *Shear Stress*  $\tau$ . The shear stress is the component of stress tangent to the plane of reference.

(q) *Stress Intensity*  $S$ . The stress intensity is defined as twice the maximum shear stress. In other words, the stress intensity is the difference between the algebraically largest principal stress and the algebraically smallest principal stress at a given point. Tension stresses are considered positive and compression stresses are considered negative.

(r) *Thermal Stress*. A self-balancing stress produced by a nonuniform distribution of temperature or by differing thermal coefficients of expansion. Thermal stress is developed in a solid body whenever a volume of material is prevented from assuming the size and shape that it normally should under a change in temperature. For the purpose of establishing allowable stresses, two types of thermal stress are recognized, depending on the volume or area in which distortion takes place, as follows:

(1) *general thermal stress*, which is associated with distortion of the structure in which it occurs. If a stress of this type, neglecting local stress concentrations, exceeds twice the yield strength of the material, the elastic analysis may be invalid and successive thermal cycles may produce incremental distortion. Therefore,

this type is classified as secondary stress in Fig. KD-230. Examples of general thermal stress are:

(a) stress produced by an axial temperature gradient in a cylindrical shell;

(b) stress produced by temperature differences between a nozzle and the shell to which it is attached;

(c) the equivalent linear stress produced by the radial temperature gradient in a cylindrical shell. Equivalent linear stress is defined as the linear stress distribution which has the same net bending moment as the actual stress distribution;

(2) *local thermal stress*, which is associated with almost complete suppression of the differential expansion and thus produces no significant distortion. Such stresses shall be considered only from the fatigue standpoint and are therefore classified as peak stresses in Fig. KD-230. Examples of local thermal stress are:

(a) the stress in a small hot spot in a vessel wall;

(b) the difference between the actual stress and the equivalent linear stress resulting from a radial temperature distribution in a cylindrical shell;

(c) the thermal stress in a liner material that has a coefficient of expansion different from that of the base metal.

(s) *Stress Cycle*. A stress cycle is a condition in which the alternating stress difference (see Article KD-3) goes from an initial value through an algebraic maximum value and an algebraic minimum value, and then returns to the initial value. A single operational cycle may result in one or more stress cycles.

## KD-220 DERIVATION OF STRESS INTENSITIES

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This paragraph describes the procedure for the linear elastic calculation of the stress intensities that are subject to specified limits. The steps in the procedure are given in the following subparagraphs. The Designer shall use the elastic-plastic analysis method (see KD-240) for cylindrical and spherical shells that have diameter ratios greater than 1.5 (see KD-251).

(a) At the point on the vessel that is being investigated, choose an orthogonal set of coordinates, such as tangential, longitudinal, and radial, and designate them the subscripts  $t$ ,  $l$ , and  $r$ . The stress components in these directions are then designated  $\sigma_t$ ,  $\sigma_l$ , and  $\sigma_r$  for direct (normal) stresses, and  $\tau_{tb}$ ,  $\tau_{tr}$ , and  $\tau_{rl}$  for shear stresses.

(b) Calculate the stress components for each type of loading to which the part will be subjected and assign each set of stress values to one or a group of the categories below (see Fig. KD-230):

- (1) general primary membrane stress  $P_m$
- (2) local primary membrane stress  $P_L$
- (3) primary bending stress  $P_b$
- (4) secondary stress  $Q$
- (5) peak stress  $F$

(c) Use Fig. KD-230 to provide assistance in assigning the stress values to the appropriate category. Consider each box in Fig. KD-230. For each of the three normal stress components and each of the three shear components, calculate the algebraic sum of the values which results from the different types of loadings. The result is a set of six stress components in each box.

(d) Translate the stress components in the  $t$ ,  $l$ , and  $r$  directions into principal stresses  $\sigma_1$ ,  $\sigma_2$ , and  $\sigma_3$ . (In many pressure vessel calculations, the  $t$ ,  $l$ , and  $r$  directions may be so chosen that the shearing stress components are zero and  $\sigma_1$ ,  $\sigma_2$ , and  $\sigma_3$  are identical to  $\sigma_t$ ,  $\sigma_l$ , and  $\sigma_r$ , respectively.)

(e) Calculate the stress differences  $S_{12}$ ,  $S_{23}$ , and  $S_{31}$  from Eqs. (1) through (3):

$$S_{12} = \sigma_1 - \sigma_2 \quad (1)$$

$$S_{23} = \sigma_2 - \sigma_3 \quad (2)$$

$$S_{31} = \sigma_3 - \sigma_1 \quad (3)$$

The stress intensity  $S$  is the largest absolute value of  $S_{12}$ ,  $S_{23}$ , and  $S_{31}$ .

Membrane stress intensity is derived from the stress components averaged across the thickness of the section. The averaging shall be performed at the stress component level in KD-220(b) or (c).

### A03 KD-230 STRESS LIMITS

Designs shall satisfy applicable requirements for the most severe combination of loadings and limits of stress intensities. For all components except spherical and cylindrical monobloc shells remote from discontinuities, those requirements are as specific in Fig. KD-230 and the subparagraphs that follow. The stress intensity calculated per KD-220(e) from the stress components in any box in Fig. KD-230 shall not exceed the stress limits that are shown in the circle adjacent to each stress intensity category in Fig. KD-230. Designs shall

include the load combinations and maximum stress limits under the conditions of design and operation.

The yield strength values  $S_y$  used in the calculations shall be taken from the tables in Section II, Part D for the design temperature, except as indicated in Note (3) of Fig. KD-230.

### KD-231 Primary Membrane and Bending Stresses

(a) The general primary membrane stress intensity  $P_m$  across the thickness of the section under consideration, due to any specified combination of design pressure and mechanical loading, shall not exceed the design stress intensity value  $kS_y/1.5$ .

(b) The local primary membrane stress intensity  $P_L$  (see Fig. KD-230) due to any specified combination of design pressure and mechanical loadings shall not exceed  $kS_y$ . The distance over which this stress intensity exceeds  $0.73(kS_y)$  shall not extend in the meridional direction more than  $\sqrt{R_m t}$ , where  $R_m$  is the radius at the midsurface of the shell or head at the location of the highest local primary membrane stress intensity measured normal to the surface from the axis of revolution in the meridional plane, and  $t$  is the nominal thickness of the shell or head under consideration.

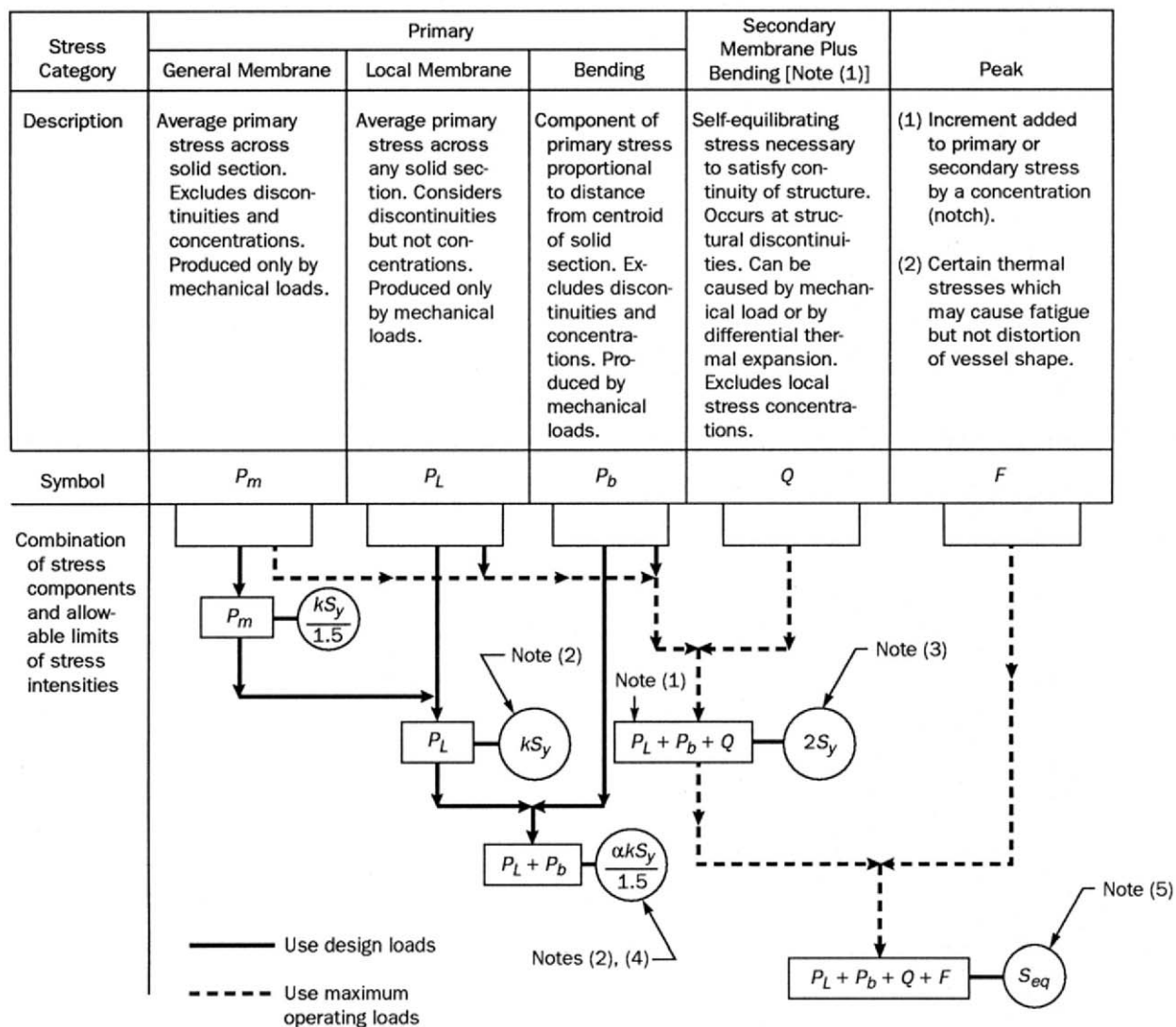
(c) The primary bending  $P_b$  plus local primary membrane stress intensity  $P_L$  due to any specified combination of design pressure and mechanical loadings expected to occur simultaneously shall not exceed  $akS_y/1.5$ . (See Fig. KD-230 when the design of components involves combinations of calculated stresses.)

### KD-232 Pure Shear Stress

The average primary shear stress across a section loaded in pure shear (for example, keys, shear rings, pins) shall be limited to  $0.4S_y$ . For screw threads, see KD-611. The maximum primary shear under design conditions, exclusive of stress concentration at the periphery of a solid circular section in torsion, shall be limited to  $0.57S_y$ .

### KD-233 Bearing Stress

(a) The average bearing stress for resistance to crushing under the maximum design load shall be limited to the yield strength  $S_y$  at temperature, except that when the distance to a free edge is greater than the distance over which the bearing load is applied, a stress of  $1.5S_y$  at temperature is permitted.



[Notes to figure follow on next page]

**A02 NOTES TO FIG. KD-230:**

GENERAL NOTE: The stress limits permitted in these design criteria are such that the maximum stress calculated on an elastic basis may exceed the yield strength of the material. The limit on primary-plus-secondary stress intensity of  $2S_y$  has been placed at a level that ensures shakedown to elastic action after a few repetitions of the stress cycle, except in regions containing significant local structural discontinuities or local thermal stresses. These last two factors are considered only in the performance of a fatigue evaluation.

**NOTES:**

- (1) The stresses in Category  $Q$  are those parts of the total stress that are produced by thermal gradients, structural discontinuities, etc., and do not include primary stresses which may also exist at the same point. It should be noted, however, that a detailed stress analysis frequently gives the combination of primary and secondary stresses directly and, when appropriate, this calculated value represents the total of  $P_m$  (or  $P_t$ ) +  $P_b$  +  $Q$  and not  $Q$  alone. Similarly, if the stress in Category  $F$  is produced by a stress concentration, the quantity  $F$  is the additional stress produced by the notch, over and above the nominal stress. For example, if a plate has a nominal stress intensity  $S$  and has a notch with a stress concentration factor  $K$ , then  $P_m = S$ ,  $P_b = 0$ ,  $Q = 0$ ,  $F = P_m(K - 1)$ , and the peak stress intensity equals  $P_m + P_m(K - 1) = KP_m$ .
- (2) The  $k$  factors shall be as follows:
  - (a) If the ratio of the minimum specified yield strength to the minimum specified tensile strength for the material is equal to or less than 0.7, then  $k = 1.0$ .
  - (b) If the ratio of the minimum specified yield strength to the minimum specified tensile strength for the material is greater than 0.7:

$$k = \frac{1 + \left[ \left( 0.5 - 0.5 \frac{S_y}{S_u} \right) / 0.3 \right]}{1.5}$$

- (3) This limitation applies to the range of stress intensity. When the secondary stress is due to a temperature excursion at the point at which the stresses are being analyzed, the value of  $S_y$  shall be taken as the average of the  $S_y$  values tabulated in Section II, Part D for the highest and the lowest temperatures of the metal during the transient. When part or all of the secondary stress is due to mechanical load, the value of  $S_y$  shall be taken as the  $S_y$  value for the highest temperature of the metal during the transient.
- (4)  $\alpha$  is the shape factor [see KD-210(o)].
- (5) This applies only if a fatigue analysis is performed in accordance with Article KD-3. See KD-140.

(b) When bearing loads are applied on parts having free edges, such as at a protruding edge, the possibility of a shear failure shall be considered. In the case of primary stress only, the average shear stress shall be limited to  $0.4S_y$ . In the case of primary stress plus secondary stress, the average stress shall not exceed  $0.5S_y$ .

(c) When considering bearing stresses from pins in supporting members, the  $S_y$  at temperature value is applicable, except that a value of  $1.5S_y$  may be used if the pin is at least three pin diameters from the edge.

**KD-234 Secondary Stresses**

The magnitude of the primary-plus-secondary stresses is limited to  $2S_y$  per Fig. KD-230.

**KD-235 Triaxial Stresses**

The algebraic sum of the three principal stresses, including the primary, secondary, and peak stresses, shall not exceed  $2.5S_y$ .

**KD-240 ELASTIC-PLASTIC ANALYSIS****01  
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The limits on general membrane stress intensity, local membrane stress intensity, primary membrane plus primary bending stress intensity, and primary-plus-secondary membrane plus bending stress intensity from KD-230, need not be satisfied and the equations for cylindrical and spherical shells in KD-250 need not be used, if an analysis is conducted using numerical methods such as elastic-plastic finite element or finite difference analysis, and the results satisfy the requirements in KD-240(a) through (f). The material shall be assumed to be ideal elastic, perfectly plastic (nonstrain-hardening<sup>1</sup>) with a yield strength equal to the value at temperature from the appropriate table in Section II, Part D.

(a) The design margin against collapse load [see KD-210(e)(6)] shall be at least 2.0 in the case of the most severe combination of specified loading.

(b) In addition, a design margin of at least 2.0 shall be applied to the higher of the following:

<sup>1</sup> A small amount of strain hardening may be used if necessary to achieve stability in the numerical solution. This strain hardening shall not result in a calculated value of stress at any point, at the maximum load applied during the analysis, which is more than 5% above the yield strength.

(1) the load that causes the strain to exceed 5% at any point in the structure. The designer shall consider the need to reduce this strain limit for confined liners made from materials which are not permitted for the primary pressure boundary in Part KM, and for areas of high triaxial tension;

(2) the load that causes yielding through the entire thickness of a cross section.

(c) Application of no more than two cycles of the maximum operating load, after application of the expected hydrotest or autofrettage load, shall result in shakedown to elastic conditions, except in small areas associated with local stress (strain) concentrations. These small areas shall exhibit a stable hysteresis loop [see KD-210(e)(9)], with no indication of progressive distortion.

(d) A fatigue analysis shall be conducted in accordance with Article KD-3, or a fracture mechanics evaluation in accordance with Article KD-4, whichever is applicable. The stress and strain values used in these fatigue analyses shall be obtained from the numerical analysis.

(e) The designer shall consider the effect of component displacements on the performance of vessel components and sealing elements, under design, hydrotest, and autofrettage loads as appropriate.

(f) The algebraic sum of the principal stresses shall not exceed  $2.5S_y$ .

**KD-251.1 Cylindrical Monobloc Shells.** The design pressure  $P_D$  shall not exceed the limit set by the equation:

$$P_D = \frac{1}{1.732} (S_y) \ln(Y)$$

**KD-251.2 Cylindrical Layered Shells.** For shells consisting of  $n$  layers with different yield strengths, the equation in KD-251.1 is replaced by:

$$P_D = \sum_{j=1}^n \frac{1}{1.732} (S_{yj}) \ln(Y_j)$$

where  $S_{yj}$  and  $Y_j$  are the yield strength and diameter ratio for each layer.

**KD-251.3 Spherical Monobloc Shells.** The design pressure  $P_D$  shall not exceed the limit set by the equation:

$$P_D = \frac{2}{1.732} (S_y) \ln(Y)$$

**KD-251.4 Spherical Layered Shells.** For shells consisting of  $n$  layers with different yield strengths, the equation in KD-251.3 is replaced by:

$$P_D = \sum_{j=1}^n \frac{2}{1.732} (S_{yj}) \ln(Y_j)$$

**KD-251.5 Additional Loads.** If the shell is subject to loading in addition to the internal pressure, the design shall be modified as necessary so that the collapse pressure in the presence of the additional load is greater than or equal to 1.732 times the design pressure.

## **KD-252 Shells Under External Pressure**

The shells shall have a diameter ratio that shall ensure the same safety against collapse as in KD-251. This means that the same equations are applicable for controlling the diameter ratios for shells under external

### **A03 KD-250 EQUATIONS FOR CYLINDRICAL AND SPHERICAL SHELLS**

Below are equations for the limits of the design pressure. The purpose of these requirements is to ensure adequate safety against collapse. These equations are only appropriate for spherical and cylindrical shells remote from discontinuities.

### **A03 KD-251 Shells Under Internal Pressure**

The shell shall have a diameter ratio  $Y$  (see KD-260) which meets the requirements of KD-251.1, KD-251.2, KD-251.3, or KD-251.4, as applicable.

pressure as those given in KD-251 for shells under internal pressure, provided that all loadings are considered and the longitudinal stress remains the intermediate principal stress (i.e., closed end case). Shells under external pressure shall also be checked for safety against buckling. For the special case of cylindrical monobloc shells, the following equation shall be used:

$$P_D = \frac{E(Y-1)^3}{40(1-\nu^2)Y^3}$$

but in no case shall exceed the value of  $P_D$  given by the equation in KD-251.1.

#### **KD-260      PRINCIPAL STRESSES IN MONOBLOC VESSELS**

For a vessel consisting of a hollow, circular cylinder of uniform wall thickness, the stresses in the cylinder wall remote from the ends of the cylinder or other discontinuities shall be calculated as follows if no residual stresses exist:

$$\sigma_t = \frac{P}{Y^2 - 1} (1 + Z^2) \quad (1)$$

$$\sigma_r = \frac{P}{Y^2 - 1} (1 - Z^2) \quad (2)$$

If the cylinder is a closed end, then:

$$\sigma_l = \frac{P}{Y^2 - 1} \quad (3)$$

where

$P$  = internal pressure

$Y$  = diameter ratio  $D_O/D_I$

$Z$  = diameter ratio  $D_O/D$

$D$  = diameter corresponding to any point in the wall

If the longitudinal stress is the intermediate principal stress, the maximum value of stress intensity  $S$  calculated from Eqs. (1), (2), and (3) is at the bore surface and is given by

$$S = P \frac{2Y^2}{Y^2 - 1} \quad (4)$$

When the longitudinal stress is not the intermediate principal stress,  $S$  shall be suitably calculated.